

CLIPMIND AI – FRONTEND DEVELOPMENT DOCUMENTATION

Name: Galigudem Harika

Role: Frontend Development

Team Details:

Team Number:4

Team Members: Galigudem Harika, Abhigna Parupalli, Tejashwini GM, AmritaPal, VijayaLakshmi, Nitesh

Start date: 19-08-2026

Internship Duration: 8 Weeks

1. Project Title:

ClipMind AI: Video Summarization & KeyMoments Detection Platform

2. Project Objective:

The main objective of ClipMind AI is to help users understand long video content quickly and efficiently without having to watch the entire video. The system processes uploaded videos, converts speech into text, generates meaningful summaries, identifies important moments, extracts keywords and highlights, and provides analytics and content insights. The project aims to reduce the time and effort required to find useful information from lengthy videos and make video content easier to understand for learners, educators, content creators, and other users.

3. Project description in detail:

ClipMind AI is an AI-based platform that helps users understand long videos easily. The user first uploads a video through the website. The system processes the video and extracts the audio. Whisper is used to convert the speech into text and create a transcript with timestamps.

The transcript is then analyzed using AI and Natural Language Processing techniques. The system generates a summary, finds important moments, extracts keywords, and creates highlights. These results are shown to the user through the application.

The frontend is developed using React.js and Vite. The backend is developed using FastAPI, and MongoDB is used to store the data. Tools such as FFmpeg and Whisper are used for video and speech processing. NLP techniques, embeddings, similarity calculation, and importance scoring are used for intelligent video analysis.

The system provides different features such as Results, Key Moments, Analytics, Content Insights, Usage Reports, and History. These features help users find important information without watching the complete video.

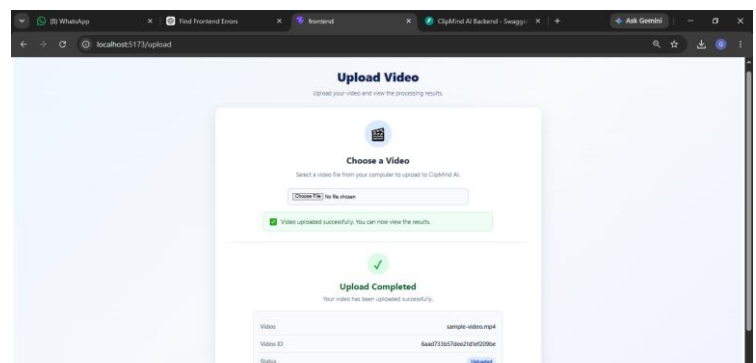
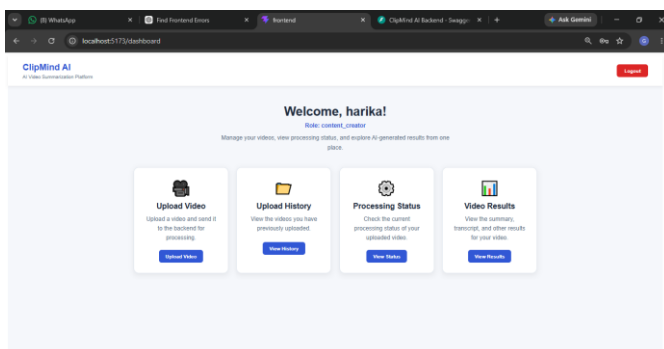
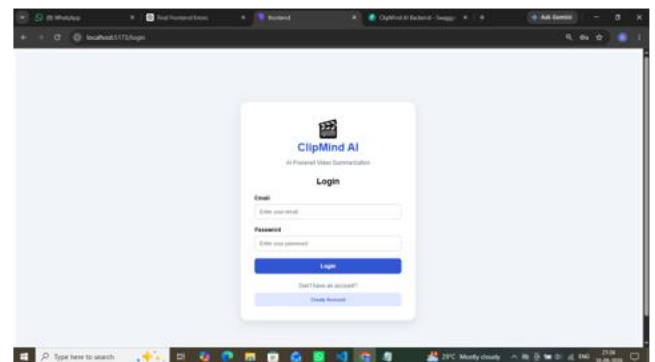
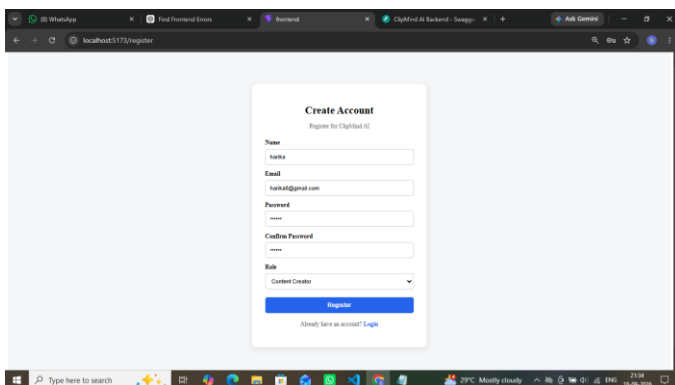
ClipMind AI can be useful in education, online learning, training, content creation, presentations, and other areas where people work with long videos. It can save time and make video information easier to understand.

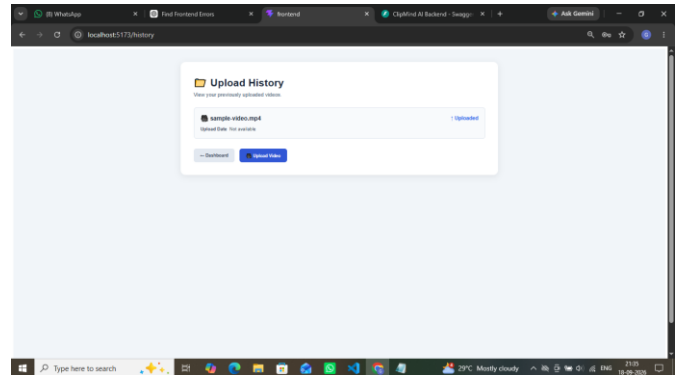
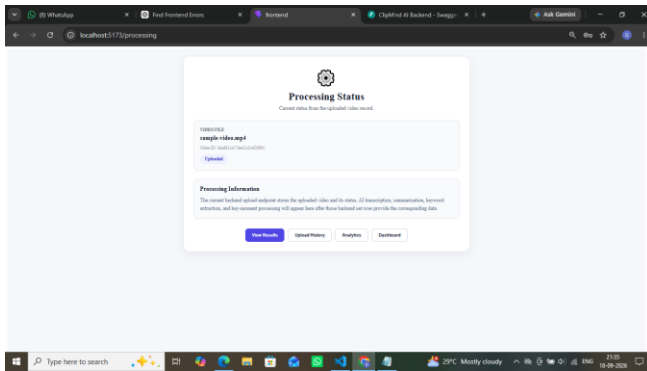
Milestone 1

In Milestone 1, the project was divided into **Module 1** and **Module 2**, and different responsibilities were assigned to different team members. The project was initially started using **Git and GitHub** for version control and team collaboration. The repository was created, branches were set up, and the team divided the development work according to individual roles.

In **Module 1**, the main activities included **Register/Login, Upload Video, and History UI**, along with the initial database setup. I was responsible for **Frontend Development and UI Planning**, where I worked on the React.js and Vite frontend and developed the Register/Login, Upload, History, Dashboard, and navigation interfaces. I also took over the **database-related work initially assigned to Vijayalakshmi**, which included working with **MongoDB and User/Video Models** for managing user and video information. Other team members worked on their assigned areas such as backend development, database support, and AI/ML-related tasks.

In **Module 2**, the focus was on the **Transcript and Summary Results UI** and storing the generated results. I continued working on the **Frontend**, where I developed the interface for displaying transcripts, timestamps, summaries, and video information. I also continued the database responsibility by working on the structure required to **store transcripts and summaries**. Other team members continued their respective responsibilities, including backend APIs, AI/ML processing, and integration.





Milestone 2:

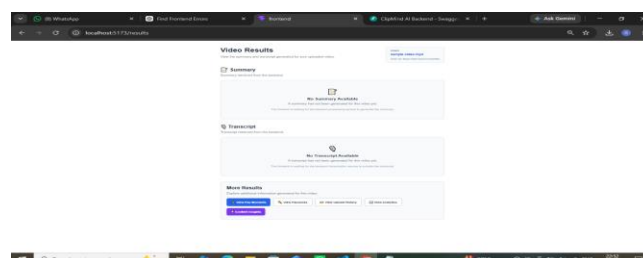
In Milestone 2, the project work was divided into **Module 3 and Module 4**. My role was **Frontend Development and Analytics Dashboard**. From **August 31 to September 5**, there was no mandatory workload for me due to examinations, and I resumed my project work on **September 7**.

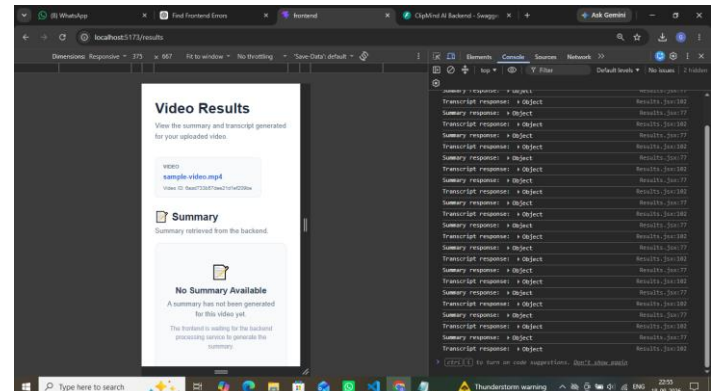
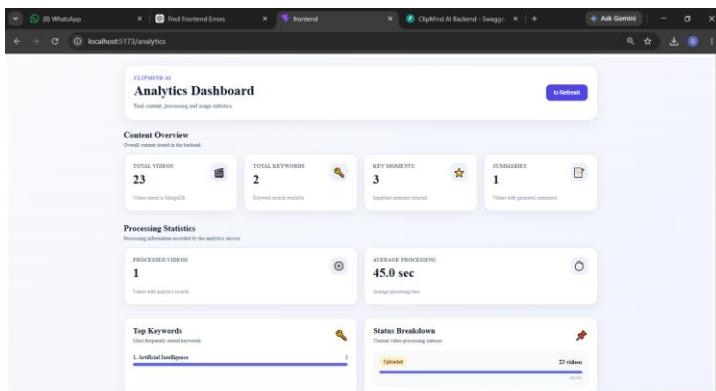
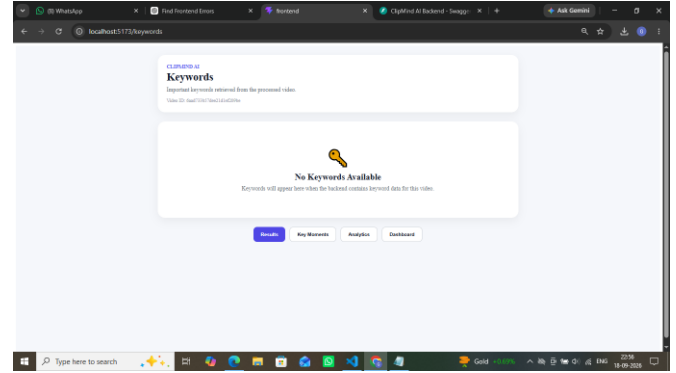
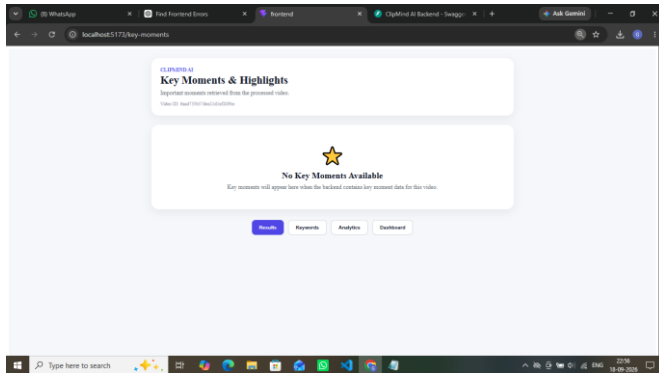
Module 3 – Key Moments, Highlights and Keywords: After resuming the project, I worked on the frontend implementation of the intelligent video analysis features. I developed the **Key Moments UI** to display important sections of the video with their timestamps. I also worked on the **highlight and keyword display**, making the AI-generated information clear and easy for users to understand. I integrated these frontend components with the backend data so that real results could be displayed instead of hardcoded content.

Module 4 – Analytics Dashboard and Frontend Integration: In Module 4, I worked on the **Analytics Dashboard** to display useful video and content statistics. I focused on dashboard layout, data presentation, responsiveness, and user-friendly navigation. I also

worked on **frontend-backend integration** so that analytics and other generated results could be displayed from the backend. Dashboard responsiveness was improved for different screen sizes, and I worked on **frontend deployment preparation**. These activities helped connect the intelligent video-processing results with the frontend and provided users with a clear visual representation of their video information.

Overall, my contribution in Milestone 2 was mainly focused on **Frontend Development and Analytics Dashboard**, including Key Moments UI, highlights, keywords, analytics, responsiveness, backend integration, and frontend deployment preparation.





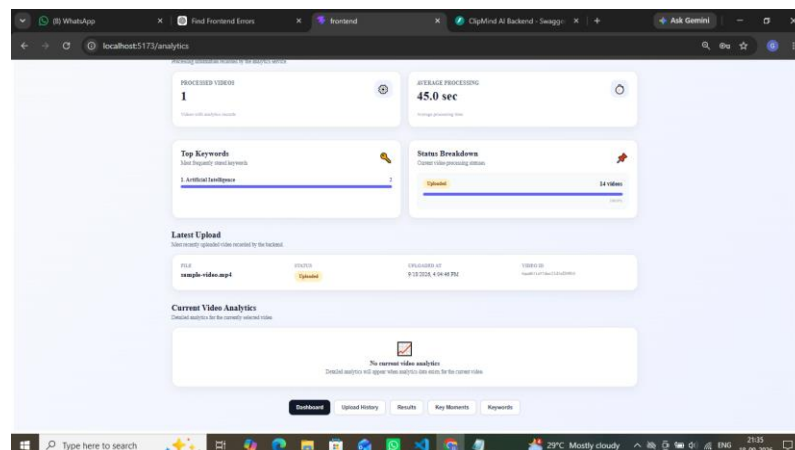
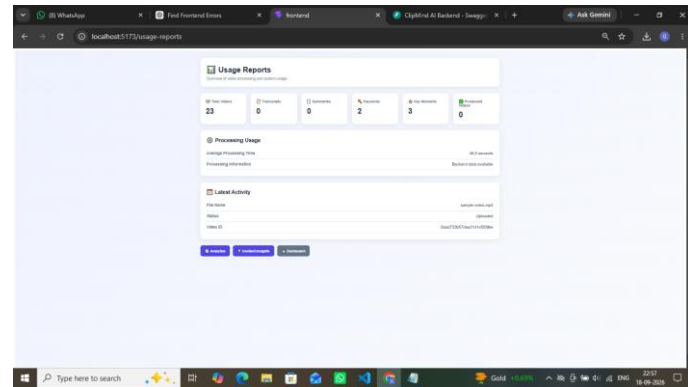
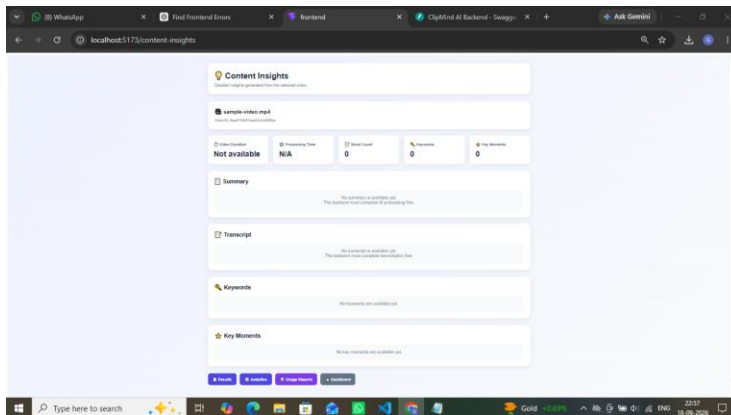
Milestone-3

In Milestone 3, the project work was divided into **Module 5** and **Module 6**, with my main role being **Frontend Development and Analytics** using React.js and Vite. The major focus was on integrating the intelligent video analysis results into the frontend, improving the user interface, and testing the complete application.

Module 5 – Intelligent Results and Analytics UI: In this module, I worked on displaying **real transcript and summary data** received from the backend. I developed and integrated the **Key Moments, Highlights, and Keywords UI**, along with timestamps to help users identify important sections of the video. I also worked on the **Analytics Dashboard, Content Insights, and Usage Statistics** interfaces. The frontend was connected with the backend APIs so that actual generated data could be displayed. A major part of my work was to **remove hardcoded and sample content** from the application and replace it with real backend data.

Module 6 – Integration, UI/UX and Testing: In this module, I focused on **frontend/backend integration, UI/UX improvements, responsive design, and frontend testing**. I checked whether the different frontend pages were correctly receiving and displaying backend data. I improved the responsiveness of the dashboard and other pages for different screen sizes and worked on loading, error, and empty states. I also tested the frontend to identify and fix interface and integration issues. The final frontend was checked to ensure that the application used real data instead of hardcoded or sample content and that the major features worked together correctly. Overall, Modules 5 and 6

helped complete the frontend integration and intelligent analysis interfaces of ClipMind AI. My contribution mainly covered **React/Vite frontend development, Key Moments, Highlights, Keywords, timestamps, Analytics Dashboard, Content Insights, Usage Statistics, backend integration, UI/UX, responsive design, removal of hardcoded content, and frontend testing.**



Milestone-4

In Milestone 4, our team decided to add new features to enhance the ClipMind AI platform. My main responsibility was the frontend implementation of the ClipMind Video DNA feature, which provides a visual fingerprint of the structure and important information in a video.

I developed the Video DNA frontend using React.js, including the visual representation of video structure, topic distribution, information density, and important-moment density. I also implemented clickable timestamps to help users identify and navigate to important sections of the video.

I connected the frontend with the `/video-dna/{video_id}` backend API and updated the required frontend and backend routing. I also focused on making the interface clear, responsive, and user-friendly. The frontend implementation is mostly completed.

Milestone	Work
Milestone 1	Git/GitHub setup, React/Vite frontend foundation, Login/Register, Upload, History, Dashboard, database-related work, Transcript & Summary UI
Milestone 2	Key Moments UI, Highlights, Keywords, timestamps, Analytics Dashboard, frontend-backend integration, responsive UI
Milestone 3	Real transcript/summary display, Key Moments/Highlights/Keywords integration, Analytics, Content Insights, Usage Statistics, removal of hardcoded data, frontend testing
Milestone 4	ClipMind Video DNA — visual video structure, topic distribution, information density, important-moment density, clickable timestamps, <code>/video-dna/{video_id}</code> API integration, frontend/backend routing

Overall Conclusion

From Milestone 1 to Milestone 4, I mainly worked on the frontend development, UI design, and Analytics Dashboard of ClipMind AI. I started by developing the basic React.js and Vite pages such as Register, Login, Dashboard, Video Upload, History, Transcript, and Summary. Later, I worked on the Key Moments, Highlights, Keywords, Analytics Dashboard, Content Insights, and Usage Statistics interfaces. I also connected the frontend with backend APIs, replaced hardcoded data with real backend data, improved responsive design, and performed frontend testing.

In Milestone 4, I worked on the ClipMind Video DNA frontend feature, including visual video structure, topic distribution, information density, important-moment density, clickable timestamps, and API integration.

Throughout the project, I received support and guidance from my team members, especially during integration, testing, and fixing issues. Working together helped me understand different parts of the project and complete my frontend responsibilities effectively.

Overall, these milestones helped me improve my practical skills in React.js, API integration, UI/UX, responsive design, analytics, Git/GitHub, and frontend testing while contributing to the development of ClipMind AI.

Acknowledgement

I would like to express my sincere gratitude to Infosys Springboard for providing me with the opportunity to work on the ClipMind AI project as part of the Virtual Internship 7.0.

I am thankful to our mentor and team members for their guidance, support, and cooperation throughout the project. Their help during development, integration, testing, and problem-solving helped me complete my frontend responsibilities effectively.

I would also like to thank my college and faculty members for their encouragement and support during the internship.

Finally, I am grateful to my team members for working together, sharing knowledge, and helping each other throughout the development of ClipMind AI. This internship gave me valuable practical experience and helped me improve my technical and teamwork skills.